

SENTHIL KUMAR J - INFORMATION TECHNOLOGY

Technology Development Program Associate at Optum

About the Company:

Optum, part of the UnitedHealth Group family of businesses, is a leading information and technology-enabled health services company, serving more than 132 million consumers and delivering the right care when and how it's needed. We are dedicated to evolving healthcare so everyone can have the opportunity to live their healthiest life.

About the Role:

The Technology Development Program (TDP) at Optum is an entry point into the world of health technology, providing recent graduates the opportunity to gain hands-on experience in developing, implementing, and supporting advanced technological solutions that improve healthcare services.

As a Software Engineering Associate within TDP, you'll contribute to projects across multiple domains, including artificial intelligence, data analytics, and application development, all while supporting the modernization of healthcare.

Overview of the Process:

The selection process consisted of three stages:

1. **Online Test** - Included coding and conceptual questions.
2. **HR Round** - A 15-minute virtual interview covering basics and interest in Optum.
3. **Technical Interview** - Conducted in CUIC, where technical skills and project experience were discussed in depth.

1. Online Test

Coding Round:

- **Problem Type:** Graph Traversal (BFS/DFS)
- **Problem Summary:** The task involved finding the shortest path in an unweighted graph to deliver a gift from John to Dave, given a network of people connected by direct relationships.
- **Approach:** Solved using BFS to determine the minimum number of steps required for the gift delivery.

Conceptual MCQs:

- The MCQs covered OOPs, DBMS, OS, and programming snippets, along with cognitive ability questions.

After this round, around 58 students were shortlisted.

2. HR Interview

The HR interview began with a casual introduction, where the interviewer introduced himself and asked about my background and interest in Optum. Key questions included:

- **Tell me about yourself:** I provided a brief introduction.
- **What do you know about Optum?:** I shared my knowledge of the company and its goals.
- **Relocation:** I confirmed my openness to relocation if required.
- **Why Optum?:** I discussed why I believe Optum is a good fit based on its values and work culture.

3. Technical Interview

In the technical interview, which took place in the afternoon, I covered a lot about my projects and technical skills. I started with a quick introduction about myself and my background before diving into my project experience.

Project Discussion:

- **First Project:** I talked about a summer project we worked on during my 2nd year. I explained the tech stack we used, some challenges we faced, and the collaborative side of working as a team. We used MongoDB for the project, and I shared why we chose it over PostgreSQL, especially emphasizing MongoDB's sharding feature, which allows better data distribution.
- **Next Project:** Then, we discussed another project from my resume. I walked the interviewer through the algorithms and features we implemented, using diagrams to give a clear idea of the implementation. He seemed interested in the details, which made it easy to explain our approach and the reasons behind our design choices.

Technical Questions:

- **Authentication vs. Authorization:** I gave an overview of both concepts, breaking down how each one works in a typical application.
- **Favorite Subject (Computer Networks):** Since I picked networks as my favorite, he asked about specific layers. I explained the network and data link layers and how IP and MAC addresses operate within them.
- **SQL Problem:** He gave me a query to solve, which I initially approached using subqueries. Then, I used joins to make it more efficient.
- **DSA Problem:** For the DSA question, I outlined my approach and coded the solution on paper, explaining its space and time complexity. I also pointed out an additional optimization, which he appreciated.

Learning New Tech Stacks: The interviewer was curious about how I learn new tech stacks. I shared that I usually combine team insights with my own research. For example, when I learned React, I took advice from teammates and also did my own deep dive before reaching a conclusion.

Why Optum?: Finally, we discussed why I was interested in Optum. I mentioned the points from their presentation that stood out to me—like their work in health technology and how their projects impact real people.

We wrapped up with a quick chat about the role of new technologies, including ChatGPT, in improving workflows. It was a positive way to end, and we both agreed on the power of tech in making life easier.

Outcome:

In total, 15 students were selected for the TDP Associate role at Optum.

My favourite part of the process was how they focused on understanding the candidates' approaches and thought processes.

My Placement Journey**Foundational Focus:**

In the last 6-8 months, I focused on building a strong foundation in data structures and algorithms (DSA) and gaining practical project experience for campus placements.

DSA Learning and Practice:

I spent a lot of time diving deep into DSA basics like arrays, sliding window, two pointers, linked lists, stacks, queues, graphs, and dynamic programming. Striver's A2Z sheet was a game-changer. I practiced on coding platforms to improve problem-solving skills, which helped me in the Optum online test, especially with the shortest-path problem.

Web Development:

I picked a web development course on Udemy, learned React, and built a fitness app, which was a great boost to my preparation.

Guidance from Seniors and Friends:

I had valuable conversations with seniors Mithulesh, Arunachalam, and Jovina about placement preparation. Mithulesh Anna, in particular, shared many insights that were incredibly helpful. Shoutout to my batchmates Dharsan and Gogilarasan for their support throughout this journey.

Support from Faculty:

I'd like to express my gratitude to Dr. K. Vani ma'am for her constant encouragement and positive insights during the placement season.